DMMST — Version 2 results: our loss variants compared with other models

1 Setup

We compare our model (a transformer with the survival head of the paper) trained with different losses against eight published survival models. All models are run on the same five datasets under the same protocol:

- 10 random splits of each dataset (60% train, 10% validation, 30% test); results are reported as mean (standard deviation) over the 10 splits;
- settings of every model chosen on the validation split only;
- every model scored by the same code.

Datasets.

- **METABRIC** (real, 1,904 patients, 1 event: death): breast-cancer patients with gene-expression and clinical features.
- **SUPPORT** (real, 8,873 patients, 1 event: death): seriously ill hospitalised adults with lab values and vital signs.
- **EBMT** (real, 2,279 patients, 4 events: recovery, adverse event, relapse, death): bone-marrow transplant patients with only four categorical features.
- **hsa synthetic** (synthetic, 5,000 patients, 2 events that can both occur): published simulated data (Tjandra et al. 2021).
- **DeepHit synthetic** (synthetic, 30,000 patients, 2 competing events): simulated data from the DeepHit paper (Lee et al. 2018).

SEER was planned as an additional large real dataset, but because of the updated SEER+ data-access policies we could not obtain access to it.

Baselines.

- **Our model (DMMST):** a transformer reads one token per feature; a survival head predicts a hazard per time interval (survival curve per event), an optional regression head predicts the event time.
- **SurvTRACE:** transformer survival model (2022), the closest to ours; run with the authors' code.
- **Cox:** classic statistical model; risk = baseline $\times$ exp(weighted sum of features).
- **DeepSurv:** Cox model with a neural network instead of the weighted sum.

- **RSF** (random survival forest): many decision trees, averaged.
- **PC-Hazard:** neural network with a piecewise-constant hazard (like our survival head, without a transformer).
- **DeepHit:** neural network predicting the probability of each event in each time interval.
- **DSM** (Deep Survival Machines): survival time as a mixture of a few smooth distributions.
- **MENSA:** recent model built for multi-event data.

Scores. The first two score the survival head, the last two the regression head.

- C_{td} (time-dependent C-index): does the model give higher risk to the patient whose event comes first? (higher is better; 0.5 is random, 1 is perfect).
- Brier score: how close is the predicted probability to what actually happened? It is the squared error, averaged over all patients (lower is better; always guessing 50% gives 0.25). IBS (integrated Brier score) is the Brier score averaged over the whole time range.
- MAE (mean absolute error): average distance between the predicted and the true event time, in the dataset's time unit (lower is better).
- Harrell C: the classic C-index, applied to the predicted times. Take every pair of patients where we know who had the event first (the earlier time must be an observed event, not censoring); the pair is correct if that patient also got the shorter predicted time. Harrell C is the share of correct pairs (higher is better; 0.5 is random, 1 is perfect). Example: A has the event at month 5, B at month 12; the model predicts 8 and 10 months: correct, even though both predictions are off. So Harrell C measures the *order* of the predicted times, MAE their *size*.

Losses and our variants. *Survival head:*

- L_{PCH}: the main loss; makes the predicted survival curves fit what really happened (event times and censoring).
- L_{rank}: compares patients; the patient whose event comes first should get the higher risk.
- L_{mul}: compares the events of one patient; the event that happens first should get the higher risk (multi-event data only).

Variants of the survival head:

Name	Losses
S1	L_{PCH} (survival likelihood)
S2	$L_{PCH} + L_{rank}$
S3	$L_{PCH} + L_{mul}$ (multi-event data only)
S4	$L_{PCH} + L_{rank} + L_{mul}$ (multi-event data only)
Full	the complete model: one joint model, S4 × R2 (S2 × R2 on single-event data)

L_{mul} compares different events of the same patient, so S3 and S4 do not exist on single-event data ("—" in the tables).

Regression head:

- L_{MAE}: the distance between predicted and true event time; for censored patients the true time is replaced by the best guess (R2–R4) or the patient is left out (R1).

- L_{MM} (mismatch): penalises the model when it predicts the wrong event to come first (multi-event data only).

The best guess is estimated from a Kaplan–Meier curve of the training data. Variants of the regression head:

Name	Losses
R1	L_{MAE} on observed events only (censored patients ignored)
R2	L_{MAE} with the best guess for censored patients
R3	R2 + L_{MM} (mismatch loss), observed times (multi-event data only)
R4	R2 + L_{MM}, best guesses (multi-event data only)
Joint	both heads trained together, every S × R combination ("Full" is one of them)

2 Results

Table 1: C_{td}, mean (sd) over 10 splits. Best per dataset in bold.

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours S1	0.667 (0.013)	0.615 (0.011)	0.529 (0.013)	0.849 (0.049)	0.743 (0.003)
Ours S2	0.670 (0.016)	0.620 (0.007)	0.535 (0.016)	**0.868** (0.010)	0.743 (0.004)
Ours S3	—	—	0.530 (0.016)	0.859 (0.015)	0.742 (0.003)
Ours S4	—	—	0.537 (0.015)	0.867 (0.009)	0.742 (0.004)
Ours Full	0.672 (0.021)	0.619 (0.007)	0.534 (0.017)	0.852 (0.041)	0.744 (0.004)
SurvTRACE	**0.678** (0.013)	**0.638** (0.008)	0.546 (0.009)	**0.868** (0.008)	0.755 (0.007)
DeepHit	0.677 (0.012)	0.610 (0.008)	0.525 (0.011)	0.865 (0.010)	0.737 (0.006)
RSF	0.672 (0.014)	0.636 (0.007)	**0.549** (0.009)	0.849 (0.009)	0.733 (0.005)
PC-Hazard	0.668 (0.017)	0.627 (0.008)	0.548 (0.010)	0.837 (0.008)	**0.762** (0.003)
DeepSurv	0.640 (0.030)	0.593 (0.007)	0.529 (0.014)	0.842 (0.009)	**0.762** (0.002)
MENSA	0.654 (0.017)	0.596 (0.021)	0.501 (0.019)	0.847 (0.014)	0.570 (0.074)
Cox	0.631 (0.022)	0.567 (0.006)	0.531 (0.014)	0.497 (0.013)	0.591 (0.005)
DSM	0.620 (0.065)	0.602 (0.037)	0.512 (0.013)	0.522 (0.031)	0.653 (0.104)

Table 2: Integrated Brier score (lower is better), mean (sd) over 10 splits. Best per dataset in bold.

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours S1	0.179 (0.012)	0.200 (0.004)	0.338 (0.165)	0.075 (0.007)	0.174 (0.005)
Ours S2	0.175 (0.006)	0.199 (0.004)	0.313 (0.153)	0.075 (0.003)	0.171 (0.005)
Ours S3	—	—	0.256 (0.082)	0.074 (0.003)	0.175 (0.005)
Ours S4	—	—	0.233 (0.043)	0.075 (0.003)	0.172 (0.006)
Ours Full	0.177 (0.005)	0.200 (0.003)	0.299 (0.138)	0.076 (0.005)	0.172 (0.007)
SurvTRACE	0.201 (0.033)	**0.193** (0.004)	0.242 (0.018)	**0.071** (0.003)	0.169 (0.006)
DeepHit	0.198 (0.007)	0.217 (0.004)	0.388 (0.159)	0.091 (0.002)	0.254 (0.012)
RSF	0.176 (0.007)	**0.193** (0.003)	0.224 (0.009)	0.087 (0.002)	0.185 (0.005)
PC-Hazard	**0.174** (0.006)	0.194 (0.004)	0.219 (0.010)	0.078 (0.003)	0.171 (0.005)
DeepSurv	0.176 (0.006)	0.195 (0.004)	0.220 (0.008)	0.078 (0.003)	**0.167** (0.005)
MENSA	0.189 (0.012)	0.212 (0.014)	0.331 (0.023)	0.079 (0.003)	0.223 (0.014)
Cox	0.175 (0.006)	0.198 (0.004)	**0.218** (0.009)	0.105 (0.002)	0.218 (0.006)
DSM	0.224 (0.067)	0.205 (0.008)	0.326 (0.024)	0.369 (0.049)	0.232 (0.078)

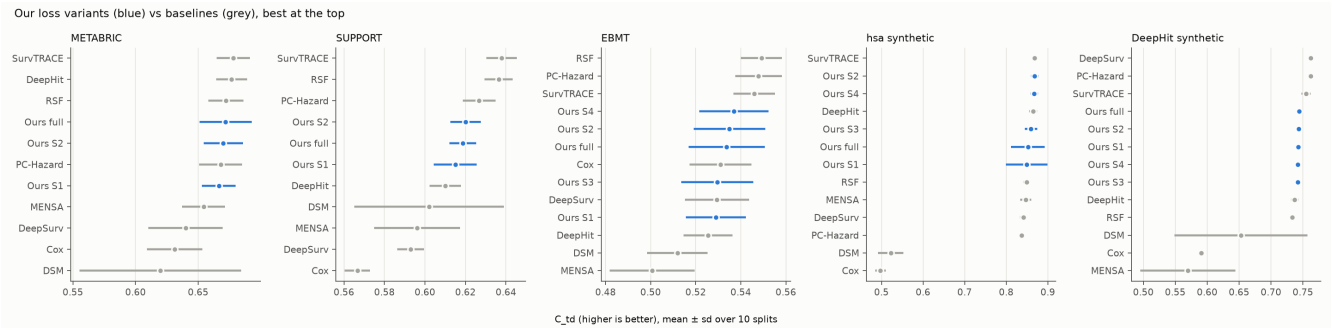


Figure 1: C_{td} of our variants (blue) and the baselines (grey), best at the top of each panel.

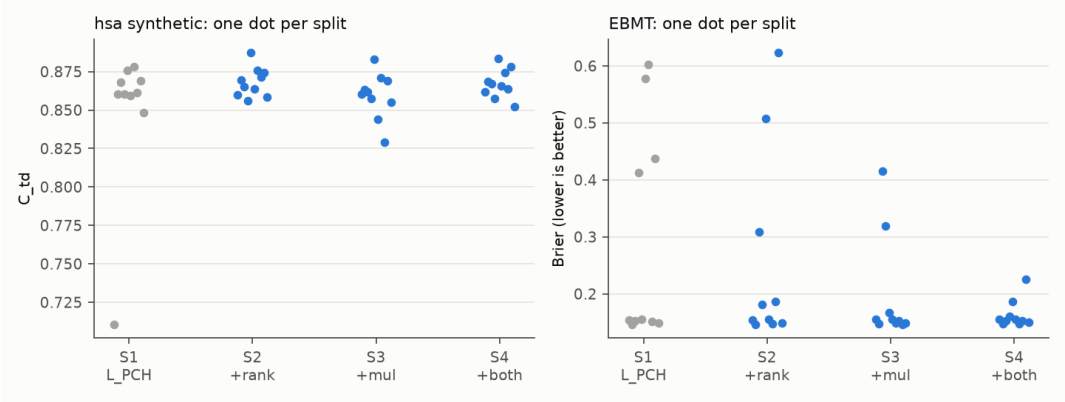


Figure 2: Every split of each loss variant. Left: C_{td} on hsa synthetic. Right: Brier score on EBMT. With L_{PCH} alone (S1) some splits fail; with $L_{rank} + L_{mul}$ (S4) none do.

Regression head

Table 3: Predicted times: Harrell C (higher is better), mean (sd) over 10 splits. Best per dataset in bold. "Full (joint)" is the same model as "Ours Full" in Tables 1–2: both heads trained together, S4 × R2 (S2 × R2 on METABRIC and SUPPORT). n/a: L_{MM} needs several events, so R3/R4 do not exist on METABRIC and SUPPORT.

Recipe	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
R1	0.608 (0.018)	0.546 (0.007)	0.510 (0.012)	0.566 (0.021)	0.689 (0.046)
R2	**0.643** (0.012)	0.592 (0.009)	0.524 (0.014)	0.663 (0.047)	0.704 (0.042)
R3	n/a	n/a	0.530 (0.009)	0.663 (0.087)	0.700 (0.035)
R4	n/a	n/a	**0.533** (0.011)	0.665 (0.063)	**0.714** (0.003)
Full (joint)	0.642 (0.015)	**0.594** (0.008)	0.505 (0.018)	**0.767** (0.112)	0.576 (0.114)

Table 4: Predicted times: MAE (lower is better), mean (sd) over 10 splits, in the time unit of each dataset (so values can only be compared within a column). n/a as in Table 3.

Recipe	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
R1	80.6 (2.8)	613.9 (9.1)	4254 (123)	396.4 (5.8)	38.8 (1.5)
R2	**76.7** (3.2)	**604.6** (9.6)	**4232** (130)	408.7 (96.7)	34.8 (3.2)
R3	n/a	n/a	4245 (126)	420.6 (94.9)	35.0 (3.0)
R4	n/a	n/a	4247 (120)	**379.8** (62.2)	**33.9** (0.4)
Full (joint)	77.6 (3.7)	609.5 (15.3)	4312 (111)	435.2 (115.8)	43.0 (5.2)

What these tables show.

- **Using the best guess for censored patients (R2) is better than ignoring them (R1).** R2 ranks predicted times better on every dataset (Harrell C 0.566 $\rightarrow$ 0.663 on hsa synthetic, 0.546 $\rightarrow$ 0.592 on SUPPORT) and has a lower MAE on four of five datasets (hsa synthetic is the exception).
- **The mismatch loss (R3, R4) adds little.** R4 is the best recipe on EBMT, hsa and DeepHit synthetic, but only by small margins over R2.
- **Joint training (Full):** on hsa synthetic training both heads together ranks times much better than the regression head alone (0.767 vs about 0.66); the best joint combination there (S1 $\times$ R4) even reaches 0.790, but it was picked by its test score, so it is optimistic. On DeepHit synthetic joint training is much worse (0.576 vs about 0.70), because joint models pick their checkpoint by the survival C_{td}, not by the quality of the predicted times. The large sd (0.112, 0.114) shows that the joint results vary a lot between splits. On METABRIC and SUPPORT, Full is as good as the regression head alone.
- **EBMT:** Harrell $C \approx 0.5$ in every setup, so predicted times there carry almost no information (the same as for the survival head).

3 Findings

1. **Our model is competitive but not the best.** It is in the upper half on every dataset and ties SurvTRACE on hsa synthetic, but wins no dataset. Our best variant is 0.000–0.018 C_{td} behind the best model.
2. L_{rank} **helps a little on every dataset** (S2 $\geq$ S1). The gain is statistically significant only on hsa synthetic.
3. **The extra losses make training reliable.** With L_{PCH} alone one hsa split collapses and several EBMT splits are badly calibrated. With $L_{rank} + L_{mul}$ (S4) this does not happen: the EBMT IBS drops from 0.338 to 0.233.
4. **Adding the regression head does not hurt survival prediction** when L_{rank} is used ("Full" $\approx$ S2/S4).
5. **For the regression head the best guess matters most** (R2 better than R1); L_{MM} adds little.
6. **Joint training helps predicted times on hsa synthetic but hurts them on DeepHit synthetic** (Harrell C 0.767 vs 0.576), because the checkpoint is chosen by the survival C_{td}, not by the regression score.
7. **Calibration is good** on four of five datasets, and better than SurvTRACE on METABRIC. EBMT is the exception.

8. **EBMT is hard for every model** (C_{td} 0.50–0.55): it has only four categorical features.

4 Conclusion

The ranking losses of the paper do not make the model more accurate than the strongest baselines, but they make it more reliable and better calibrated, and they allow the regression head to be trained together with the survival head at no cost. The results do not support a claim of better C_{td} than SurvTRACE.